

Lag-Order Uncertainty in Set-Identified SVARs

A comparative presentation of two Monte Carlo studies

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Agenda

Results

Introduction

- ▶ Inspiration from Ivanov and Kilian (2005) who compare the performance of different lag-order selection criteria in a Monte Carlo study in how well they can recover the true IRF of a VAR model.
- ▶ Since then VAR estimation has changed a lot, we aim to update their analysis to a more current framework:

Feature	Ivanov and Kilian (2005)	Our Approach
IRF	Point-Identified (Cholesky Decomposition)	Set-Identified (Sign Restrictions)
Model Selection	Information Criteria	Information Criteria vs. BVAR and BMA
Data	Macroeconomic Studies from the 1990s/early 2000s (both monthly and quarterly)	Kilian and Murphy (2012) (monthly)

Simulation Design

- ▶ Using VAR specification with true IRF specific $\tilde{B}$ to median target IRF from initial estimation on real data from Kilian and Murphy (2012).
- ▶ Estimation for lag orders $p_0 \in \{4, 10\}$, and sample sizes $T \in \{240, 360, 480, 600\}$
- ▶ 500 Monte Carlo iterations R for each combination of p_0 and T , 1000 admissible IRFs selected for each model in each iteration.
(Computational limits even with parallel processing and translation into machine code with `numba` and `joblib` packages in `python`)
- ▶ Metric for performance comparison: Geometric mean of squared errors of median IRF of model j relative to the median IRF of a model correctly specified with p_0 :

$$MSE_j^{relative} = \exp \left(\frac{1}{n} \sum_{i=1}^R \ln \left(\frac{SE_{ji}}{SE_{p_0 i}} \right) \right) \quad (1)$$

IC Lag Selection

Table: Average Evaluated Lag Order ($\hat{p}$) selected by Information Criteria.

p_0	Estimator	AIC	BIC	HQC
	T			
4	240	2.99	1.95	2.06
	360	3.39	2.00	2.16
	480	3.61	2.01	2.40
	600	3.80	2.02	2.66
10	240	6.54	1.96	2.07
	360	8.53	2.00	2.23
	480	8.96	2.00	2.66
	600	9.16	2.02	4.03

IC Performance Comparison

Table: Relative Mean Squared Error: Information Criteria Baselines.

p_0	T	AIC	BIC	HQC
4	240	1.083	1.108	1.129
	360	1.117	1.274	1.296
	480	0.986	1.133	1.124
	600	0.989	1.145	1.118
10	240	1.033	1.018	1.021
	360	0.811	0.904	0.878
	480	0.989	1.190	1.185
	600	1.036	1.260	1.259

Note: Bold values indicate the lowest Relative Mean Squared Error across the respective scenario.

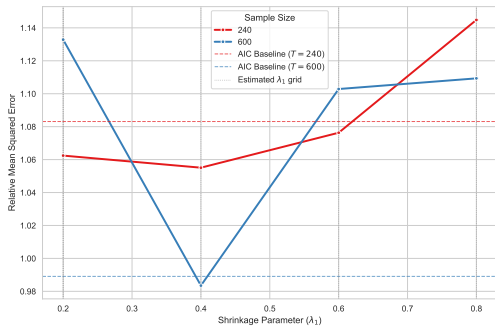
BVAR vs. IC Performance Comparison

Table: Relative Mean Squared Error of Bayesian Shrinkage Specifications vs. AIC Baseline.

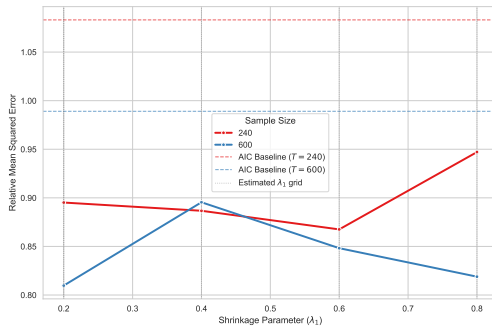
p_0	T	BVAR ($\lambda_1 = 0.20$)	BVAR ($\lambda_1 = 0.40$)	BVAR ($\lambda_1 = 0.60$)	BVAR ($\lambda_1 = 0.80$)	BVAR (MDD λ_1)
4	240	0.981	0.974	0.994	1.057	0.981
	360	0.922	1.045	0.890	1.022	0.922
	480	1.153	1.125	1.108	1.116	1.153
	600	1.145	0.994	1.115	1.122	1.145
10	240	0.866	0.858	0.840	0.917	0.866
	360	1.105	1.110	1.055	1.116	1.124
	480	0.863	0.931	0.886	0.955	0.876
	600	0.782	0.865	0.819	0.791	0.796

Note: Bold values indicate the lowest Relative Mean Squared Error across the respective scenario.

BVAR Performance Depending on λ_1



(a) $p_0 = 4$



(b) $p_0 = 10$

Figure: Comparison of BVAR's performance across different λ_1 -values in comparison to AIC.

MDD Selection of λ_1

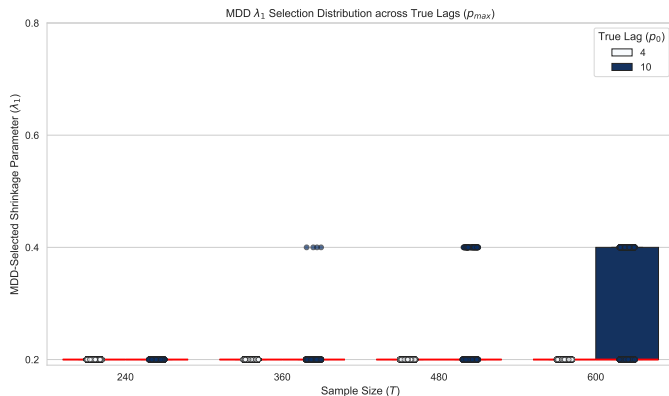


Figure: The MDD selected hyperparameter λ_1 for different sample sizes (both $p_0 = 4$ and $p_0 = 10$).

Note: Circles represent optimal λ_1 -values in individual iterations, bars represent the interquartile range, red lines indicate the median.

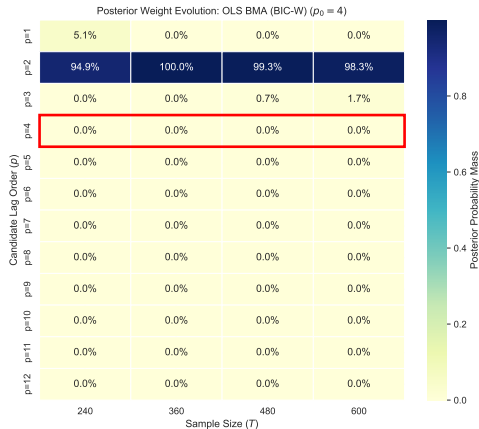
BMA vs. IC Performance Comparison

Table: Relative Mean Squared Error of Bayesian Model Averaging Specifications vs. AIC Baseline.

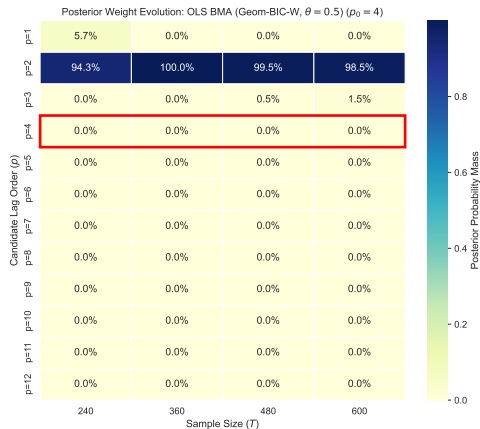
p_0	T	OLS BMA (BIC)	OLS BMA (Geom. BIC)	OLS BMA (AIC)	OLS BMA (Geom. AIC)	Joint Grid BMA	Geom. Grid BMA
4	240	1.015	1.074	0.999	1.085	0.831	1.040
	360	1.023	1.106	0.927	1.019	0.919	0.937
	480	1.164	1.166	1.046	1.010	0.911	1.147
	600	1.231	1.141	1.043	1.136	1.125	1.122
10	240	0.978	0.815	0.948	0.950	0.775	0.821
	360	1.434	1.304	1.102	1.090	0.970	1.224
	480	1.064	1.297	0.893	1.224	1.031	0.908
	600	1.248	1.062	0.852	0.845	0.847	0.930

Note: Bold values indicate the lowest Relative Mean Squared Error across the respective scenario.

BMA Weighting (OLS BMA - BIC)

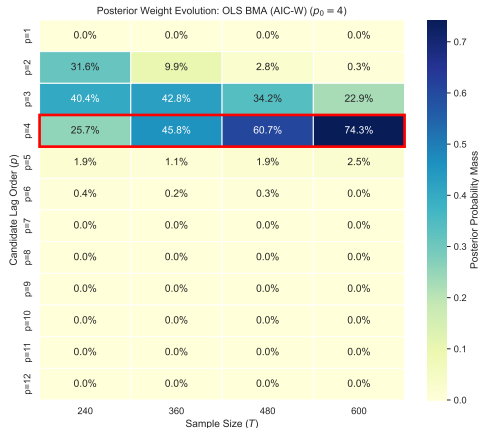


(a) OLS BMA (BIC-weighted)

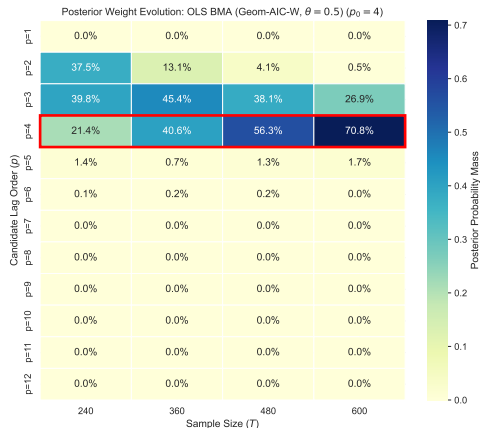


(b) OLS BMA (Geom. BIC-weighted, $\theta = 0.5$)

BMA Weighting (OLS BMA - AIC)



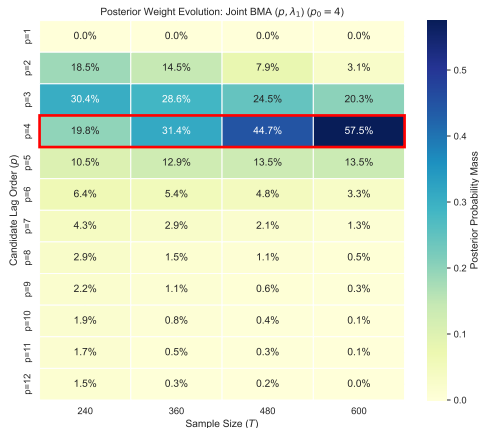
(a) OLS BMA (AIC-weighted)



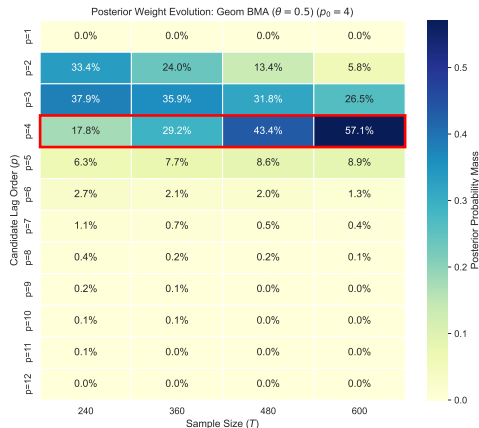
(b) OLS BMA (Geom. AIC-weighted, $\theta = 0.5$)

Figure: Posterior Weight Evolution across the four OLS BMA architectures ($p_0 = 4$).

BMA Weighting (Aggregated for both Grid Approaches)



(a) Joint Grid BMA



(b) Geometric Grid BMA

Figure: Posterior Weight Evolution for the two BVAR-based BMAs ($p_0 = 4$).

Average shrinkage changes with p

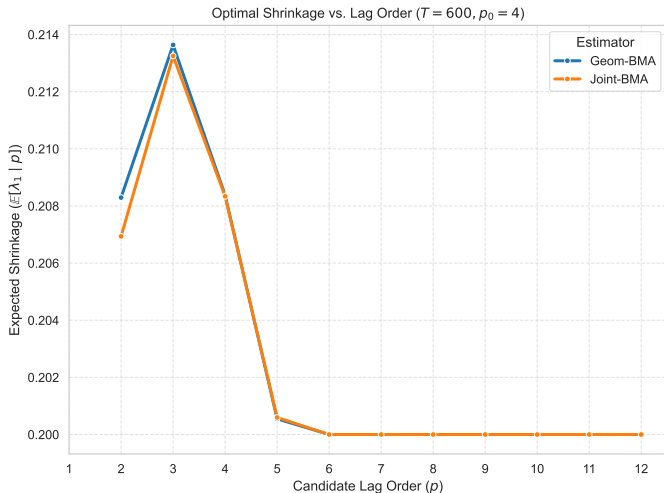
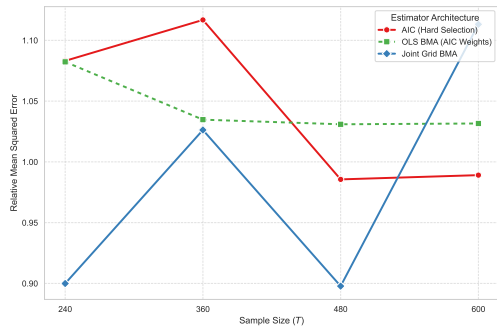
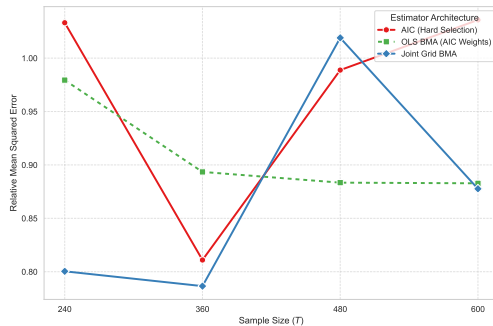


Figure: The expected weight of λ_1 given p changes with p ($p_0 = 4$).

Performance of BMA vs Hard Selection at Different Sample Sizes



(a) $p_0 = 4$



(b) $p_0 = 10$

Figure: Performance of BMA vs Hard Selection at different sample sizes T .